

Gas Agency Management System

Team Name: Pega Challengers

Team Members:

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Use Case Description: Gas Agency Management System is an online application for a gas agency that customers from different geographical regions use to book gas cylinders. It is owned and operated by a company called GAM. Gas agencies get booking requests in person or via phone calls. The main objective of this application is to provide an effective and automated gas booking procedure that is aimed to reduce the manual work for servicing customers effectively.

Planning and Design: Designed for GAM, the Gas Agency Management System (GasMS) automates cylinder bookings through three core workflows: Consumer Registration, Gas Booking, and Connection Transfer. A key design element is the dynamic pricing logic, which charges \$75 for domestic cylinders within a 12-unit annual subsidy limit and \$150 otherwise. The application enhances operational efficiency for Consumers and Managers by integrating automated email invoices, delivery tracking, and comprehensive performance reporting.

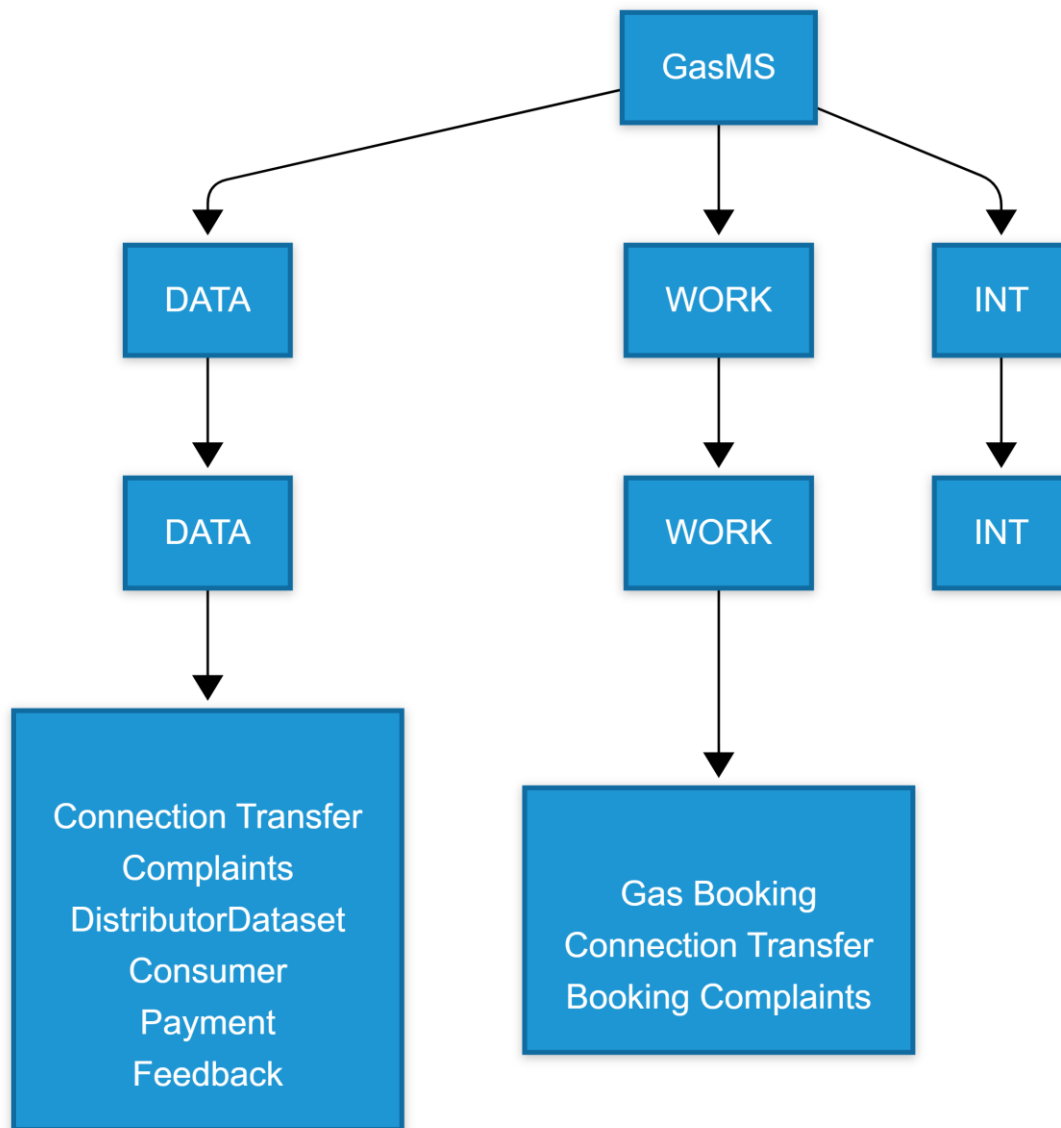
At a high level and with the initial phase I release, this application will:

- To be able to book a cylinder online, you should be a registered customer. If you are not registered, you can register online by providing details like consumer number, 17 digits LPG ID, Aadhar number, mobile number and contact address.
- After you have registered, you can log on to the Gas online portal using the registered phone no or email id.
- Once you have logged in, you will be able to book gas cylinder online easily. All you have to do is just enter the required details and submit the form to place an order.
- The order is then sent to the concerned distributor/agency and you will be able to pay to the distributor directly on delivery/ online through card or net banking. Also, you will be able to track the status of your booking online.
- Once delivery complete consumer will provide the feedback to the agency about the service.
- Agency manager will go through the consumer feedback report to modify or improve the better service.

Application Structure: Implementation

Application Name: Gas Agency Management System

ECS LAYER:

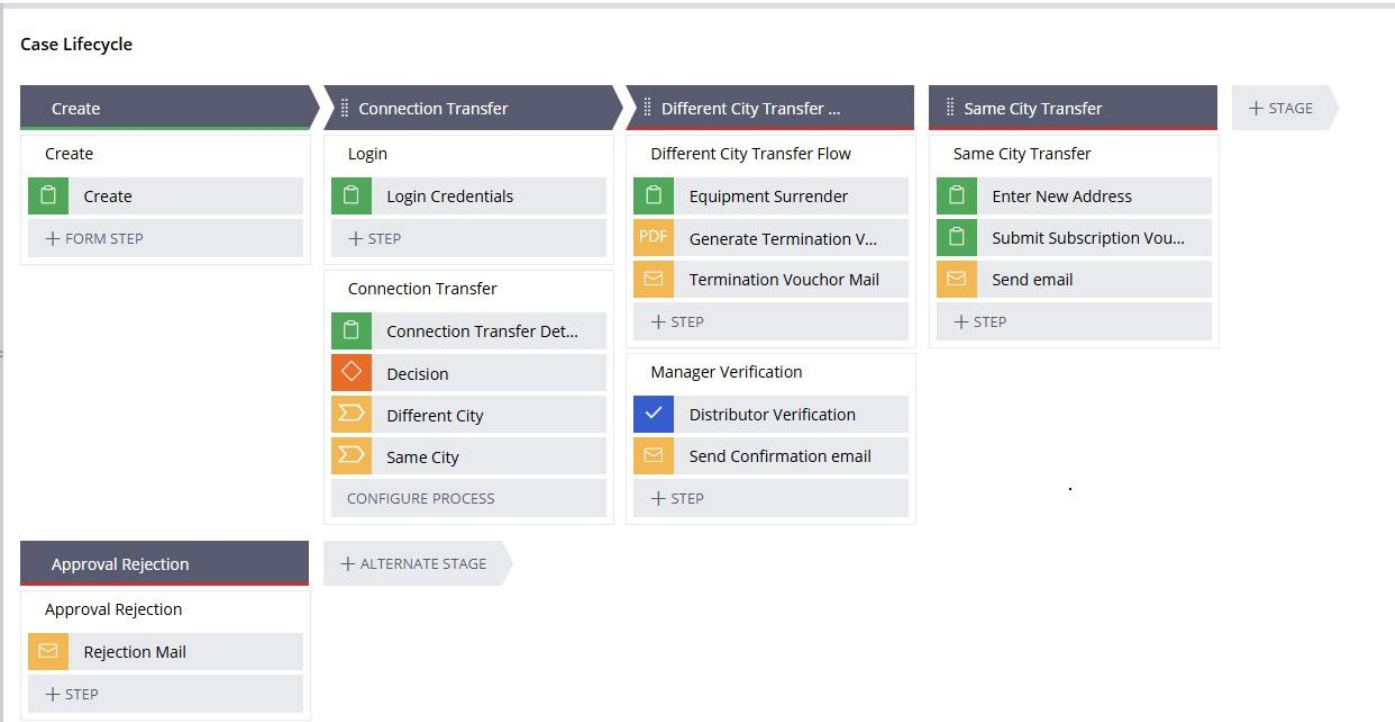


CASE TYPE USED AS FOLLOWS :

Gas Booking:



Connection Transfer:



Booking Complaints:

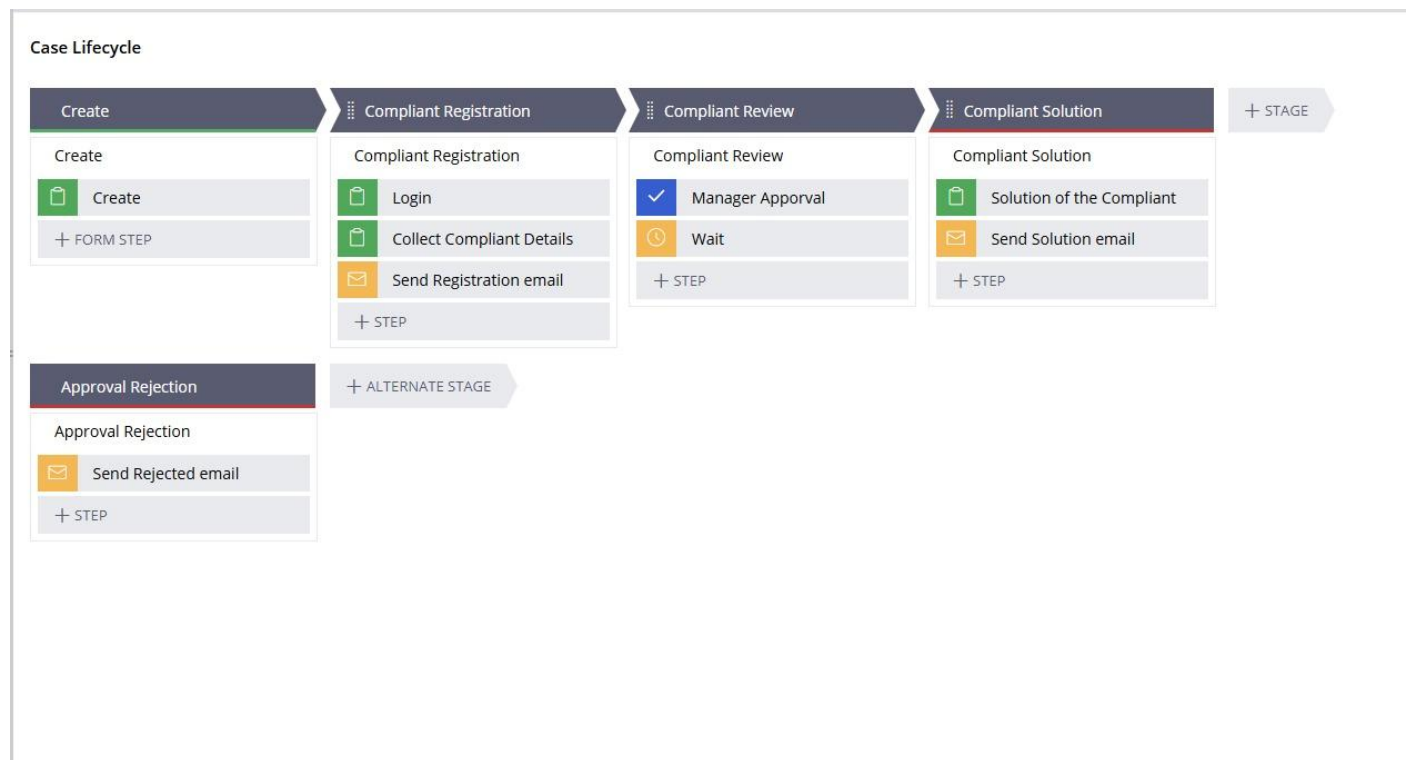
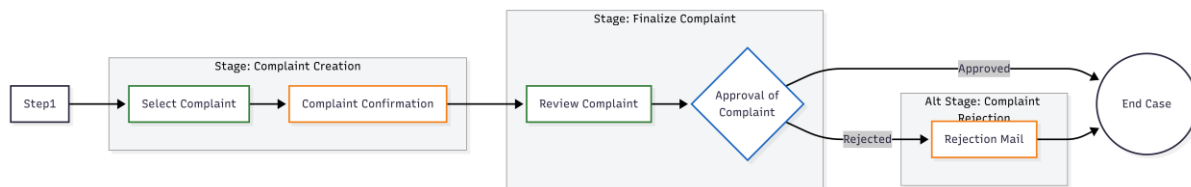


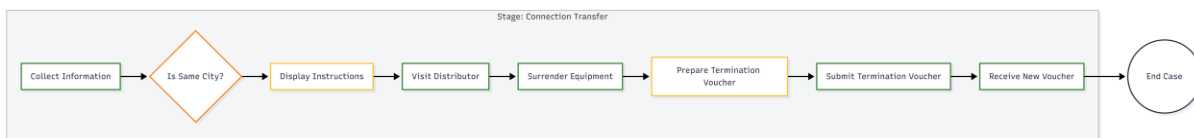
Fig: The above figures represents the case life cycle of the application.

FLOW DIAGRAM

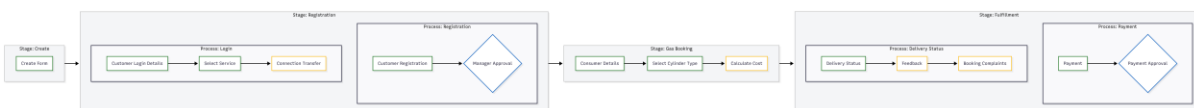
Booking Complaints:



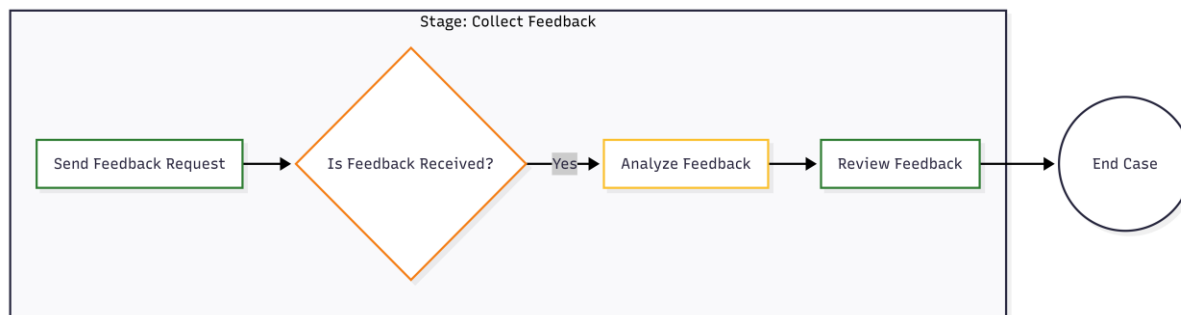
Connection Transfer:



Gas Booking:



Gas Booking:



Process Empty

DFD DIAGRAM:

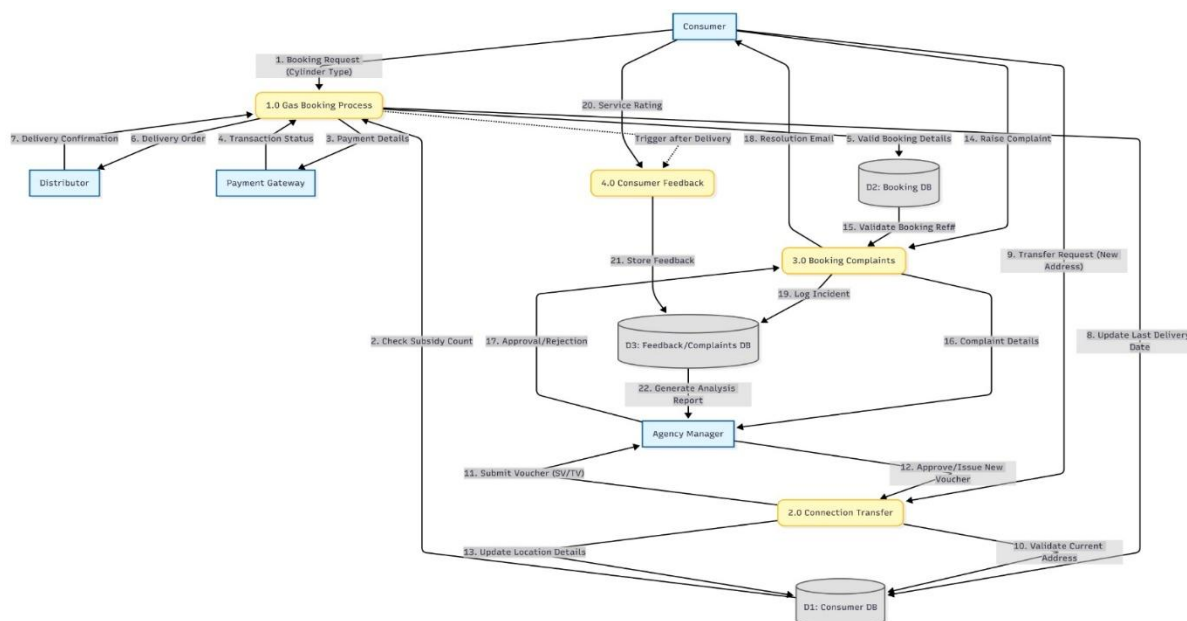


Fig: The above figure represents the data flow diagram

DATA TYPES USED :

1.Connection Transfer

Search...	Q	☐ Show system fields	View data model	Primary Fields	Add Field
Name	ID	Type	Options	Application Layer	
Address	Address	Text (single line)		Gas Agency Management	
ConsumerNumber	ConsumerNumber	Text (single line)		Gas Agency Management	
Distributor	Distributor	Text (single line)		Gas Agency Management	
EquipmentSurrendered	EquipmentSurrendered	Boolean		Gas Agency Management	
New Address	NewAddress	Text (single line)		Gas Agency Management	
TerminationVoucherNumber	TerminationVoucherNumber	Text (single line)		Gas Agency Management	
TransferRequestDate	TransferRequestDate	Date only		Gas Agency Management	
TransferType	TransferType	Picklist	Drop-down list	Gas Agency Management	
UserName	UserName	Text (single line)	Key; Read-only	Gas Agency Management	

2.Consumer

Search...

☐ Show system fields

View data model

Primary Fields

Add Field

Name	ID	Type	Options	Application Layer	
AadharNumber	AadharNumber	Text (single line)		Gas Agency Management	 
City	City	Text (single line)		Gas Agency Management	 
Confirm Password	ConfirmPassword	Text (single line)		Gas Agency Management	 
ConsumerNumber	ConsumerNumber	Text (single line)	Key; Read-only	Gas Agency Management	 
ContactAddress	ContactAddress	Text (single line)		Gas Agency Management	 
DistributorName	DistributorName	Text (single line)		Gas Agency Management	 
District	District	Text (single line)		Gas Agency Management	 
Email	Email	Email		Gas Agency Management	 
LPG_ID	LPG_ID	Text (single line)		Gas Agency Management	 
MobileNumber	MobileNumber	Phone		Gas Agency Management	 
Password	Password	Text (single line)		Gas Agency Management	 
RequiredFor	RequiredFor	Picklist	Drop-down list	Gas Agency Management	 
State	State	Text (single line)		Gas Agency Management	 
SubsidizedCylinderCount	SubsidizedCylinderCount	Integer		Gas Agency Management	 
UserName	UserName	Text (single line)		Gas Agency Management	 

3.Distributor Dataset:

Search...

☐ Show system fields

View data model

Primary Fields

Add Field

Name	ID	Type	Options	Application Layer	
City	City	Text (single line)		Gas Agency Management	 
DistributorName	DistributorName	Text (single line)		Gas Agency Management	 
District	District	Text (single line)		Gas Agency Management	 
S.No	SNo	Text (single line)	Key; Read-only	Gas Agency Management	 
State	State	Text (single line)		Gas Agency Management	 

4. Feedback:

Search...

☐ Show system fields

View data model

Primary Fields

Add Field

Name	ID	Type	Options	Application Layer	
AreasForImprovement	AreasForImprovement	Text (single line)		Gas Agency Management	 
Comments	Comments	Text (single line)		Gas Agency Management	 
DeliveredOnTime	DeliveredOnTime	Boolean		Gas Agency Management	 
Delivery Service	DeliveryService	Picklist	Radio-buttons	Gas Agency Management	 
FeedbackDate	FeedbackDate	Date only	Calculated; Read-only	Gas Agency Management	 
ID	ID	Text (single line)	Key; Read-only	Gas Agency Management	 
Rating	Rating	Text (single line)		Gas Agency Management	 
StaffBehavior	StaffBehavior	Text (single line)		Gas Agency Management	 

5. Payment Details:

Search...	Q	☐ Show system fields	View data model	Primary Fields	Add Field
Name	ID	Type	Options	Application Layer	
Card Holder Name	CardHolderName	Text (single line)		Gas Agency Management	 
Card Number	CardNumber	Text (single line)		Gas Agency Management	 
CVV Code	CVVCode	Integer		Gas Agency Management	 
Expiry Date	ExpiryDate	Date only		Gas Agency Management	 
Payment ID	PaymentID	Text (single line)	Key; Read-only	Gas Agency Management	 
Payment Type	PaymentType	Picklist	Drop-down list	Gas Agency Management	 
UPI ID	UPIID	Text (single line)		Gas Agency Management	 

6.Compliants:

Search...	Q	☐ Show system fields	View data model	Primary Fields	Add Field
Name	ID	Type	Options	Application Layer	
Booking ID	BookingID	Text (single line)	Key; Calculated; Read-only	Gas Agency Management	 
Compliant Date	CompliantDate	Date only	Calculated; Read-only	Gas Agency Management	 
Compliant Type	CompliantType	Picklist	Drop-down list	Gas Agency Management	 
Manager Notes	ManagerNotes	Text (single line)		Gas Agency Management	 
Reason	Reason	Text (single line)		Gas Agency Management	 
Solution	Solution	Text (single line)		Gas Agency Management	 

ACTORS:

- **Consumer** – Consumers can book a cylinder by providing their consumer Id and can pay online at the time of booking or on delivery of the cylinder.
- **Agency Manager** – Approves / rejects the new consumer request and approves/rejects the request for booking s new cylinder

PORTALS:

- User Portal
- Manger Portal

ACCESS GROUP :

- GasMS: Users
- GasMS: Authors
- GasMs: Managers

WORK QUEUE:

To improve workload management and task routing in your application, create a work queue that holds assignments for operators and robotic queues. Because you associate a work queue with a group of users or a robotic queue, you logically and efficiently categorize work inside your organization.

Edit Work Queue: ManagerWorkQueue

ID: ManagerWQ RS: GasAgency [Edit]

Work Queue Preferences History

General

Type

Standard

Organization

Name

GAM

Division

Div

Unit

Unit

Work group

ManagerWG

Contacts

1

Manager1GAM

2

Manager2GAM

3

Manager3GAM

4

Manager4GAM

Roles

1

GasAgency:ManagerManaged

Fig: The above figure represents the work queue.

Login Page:

U

Login Credentials

Assigned to User@GAM • In G-4002 • Urgency 10

UserName *

Enter your username

Password *

Enter password

☐ New User

Cancel

Fill form with AI

Save for later

Submit

Registration Page

SS

Customer Registration
Assigned to [Senior System Architect](#) • In G-4002 • Urgency 10

Consumer Details

UserName *
Enter Username

Email *
Enter Email ID

MobileNumber *
+1

Enter Mobile Number

ContactAddress *
Enter Contact Address

Password *
Enter Password
Password must contain at least one capital alphabet , numeric & special character

Confirm Password *
Re-enter the password

RequiredFor *
Select...

AadharNumber *
Enter Aadhar number
Aadhar must contain 12 digits

Aadhar *

Drop or choose a file

State
Select...

Cancel

✦ Fill form with AI

Save for later

Submit

OPERATORS:

- ❖ User@GasMS
- ❖ Manager@GasMs